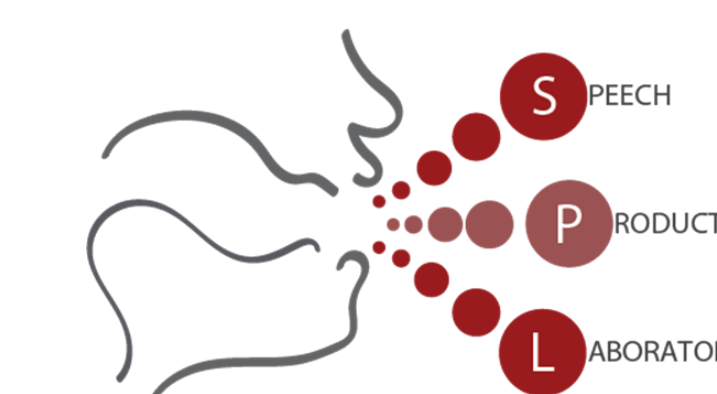




Cylindrical coordinates and tongue grooving detection in 3D/4D ultrasound

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Introduction: tongue grooving

- English liquids → multiple constrictions
 - 'Bunched' /ɹ/: labial + palatal + (lower) pharyngeal
 - 'Dark' /ɹ/: coronal + uvular
- Lindblad and Lundqvist (2003): one motor action → multiple constrictions
- Compression of medial tissue between two locations → displacement of tongue surface in both locations
- Evidence for medial compression: grooving in the tongue midline

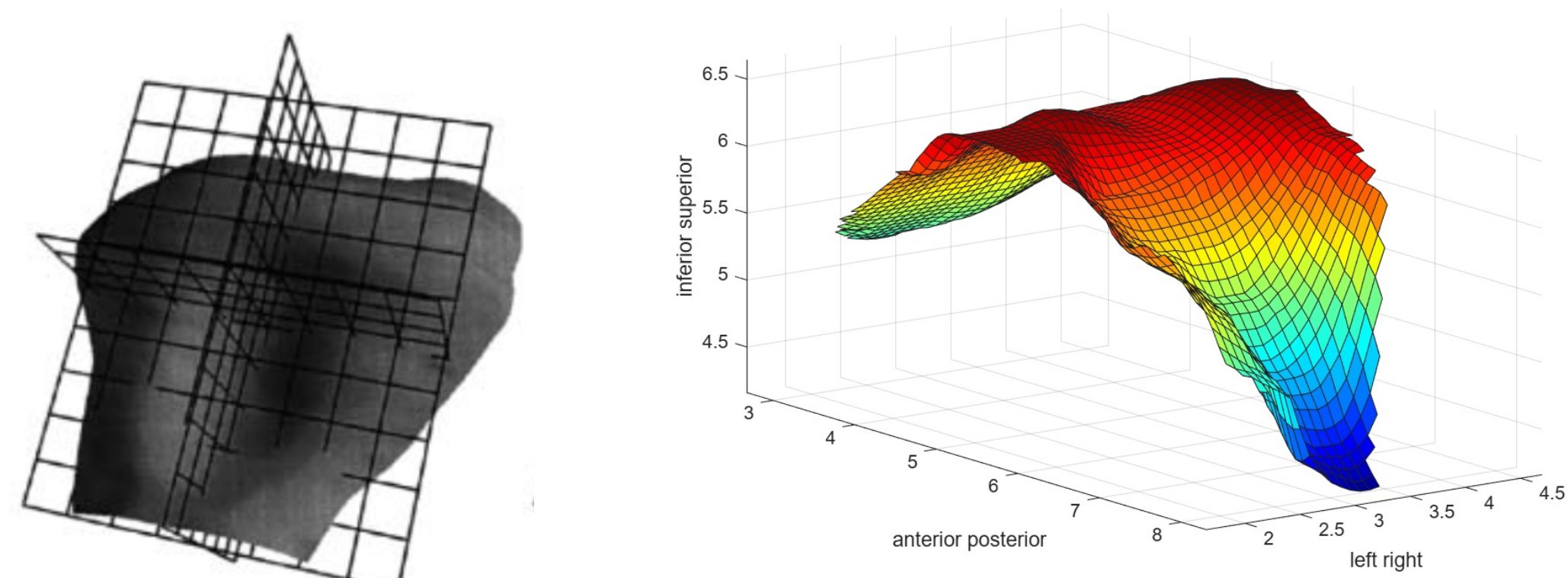


Figure 1: Left: grooved /ɹ/ from uncalibrated ultrasound data (Lundberg and Stone 1999). Right: grooved /ɹ/ from calibrated 3D/4D ultrasound

Introduction: cylindrical coordinates

- Difficult to detect grooving in the round surface of tongue in sagittal plane
- Cartesian coordinates → cylindrical (polar) coordinates
 - Need origin for polar space (Mielke 2015)
 - (x, y) are converted into radial distance (r) from origin at particular angle (θ)
- Polar coordinates approximate 2D midsagittal tongue surface as a circle
- Cylindrical coordinates approximate 3D tongue surface as a cylinder
- Region with tongue grooving will be lowest in cylindrical surface

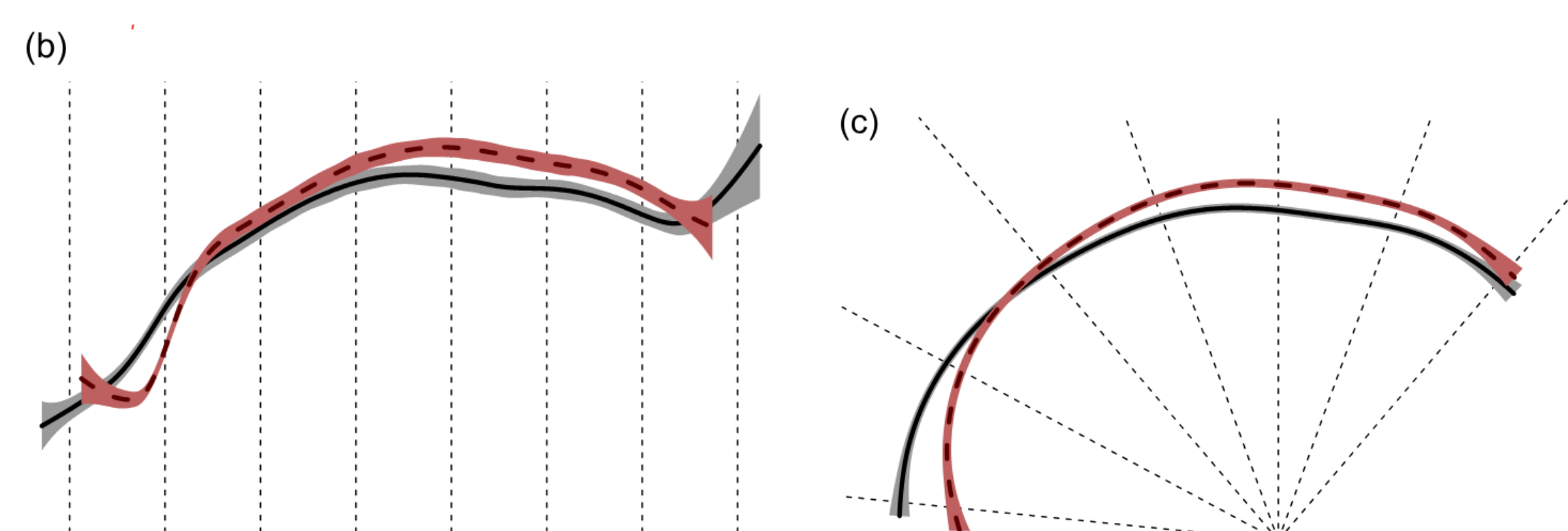


Figure 2: Cartesian coordinates (left) vs. polar coordinates (right) (from Mielke 2015). Polar coordinates allow for different regions of the tongue to be on relevant axes of variation

Methods

- 14 speakers (8 female) from Indiana, aged 18-22
- Stimuli include CVC words containing /ɹ/ and /l/ in coda and onset position, with /i/, /a/, and /ou/ nuclei (i.e., leap, peel, leer, real)
 - Carrier phrase "I said a ____ again."
- Phillips EpiQ 7G 3D/4D ultrasound system with a Phillips xMatrix x6-1 digital 3D/4D transducer (Lulich et al. 2018)
- Tongue traces were made using 3D-SLURP (Naga Karthik et al. 2020)
- Data analysis was performed in WASL (Lulich 2020) in MATLAB

Post-velar ultrasound (at LabPhon 20)

Results: determining an origin

- "search space" containing all tongue traces was created
- Every point within search space was tested as a possible origin, and variance (distance from origin and all points) was measured
- The origin with lowest variance was selected for that particular file
- Generally, more variation in the z-values of determined origins across speakers (figures 3 and 4)
- As z increases (meaning origin is closer to tongue surface), variance generally increases across speakers (figure 5)
- The mean x- and z-value of all determined origins for a particular speaker was used when converting into cylindrical coordinates

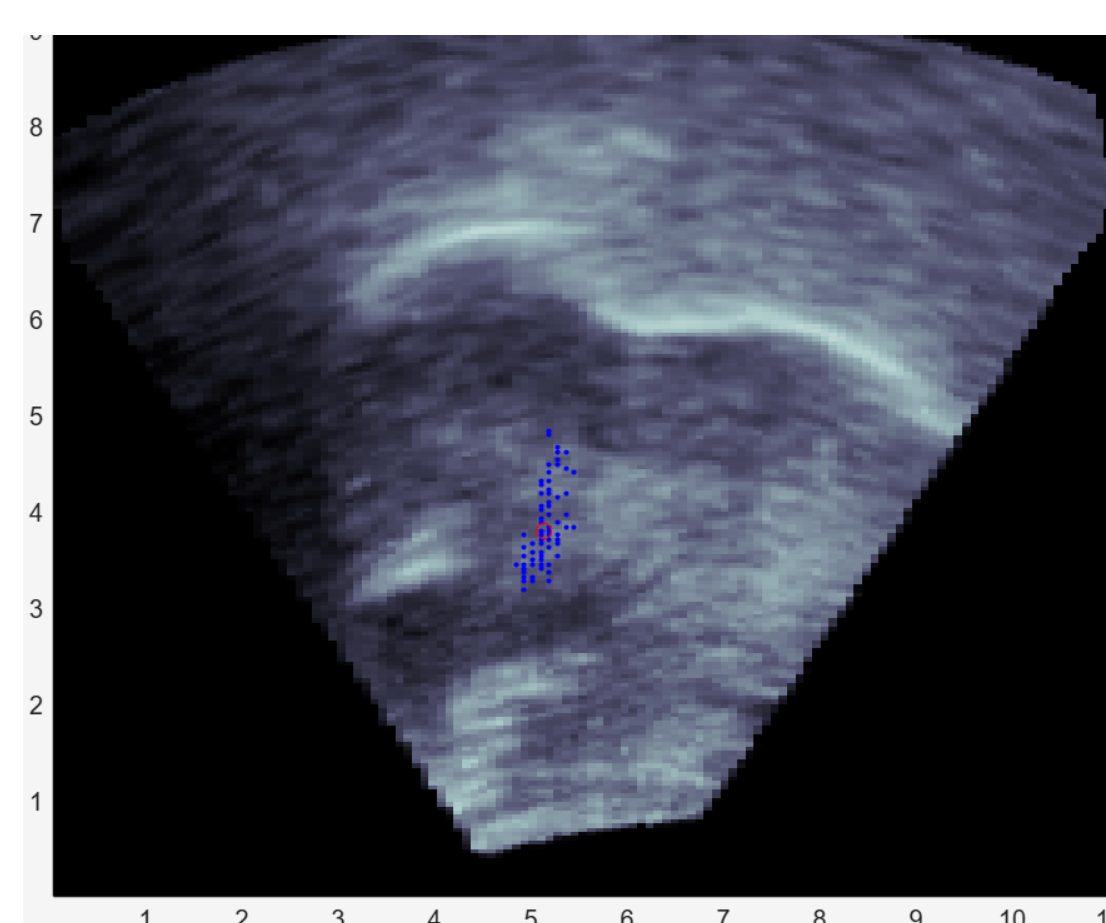


Figure 3: Origins from all files plotted on ultrasound image (blue dots) and mean origin (red circle). Ultrasound image shows the midsagittal view of an onset /ɹ/

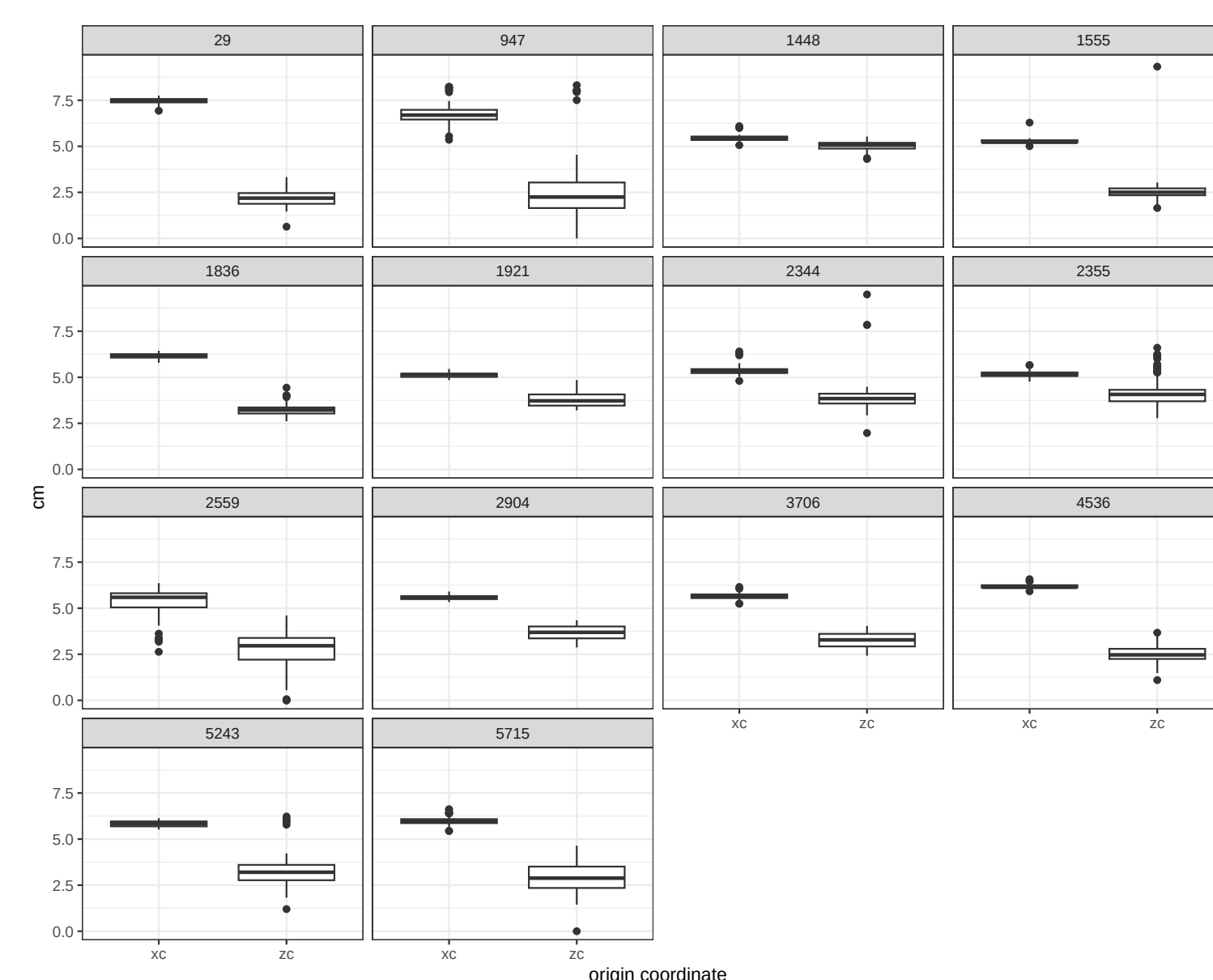


Figure 4: Distribution of x and z values of origins across speakers. There is more variation in the superior/inferior space (z)

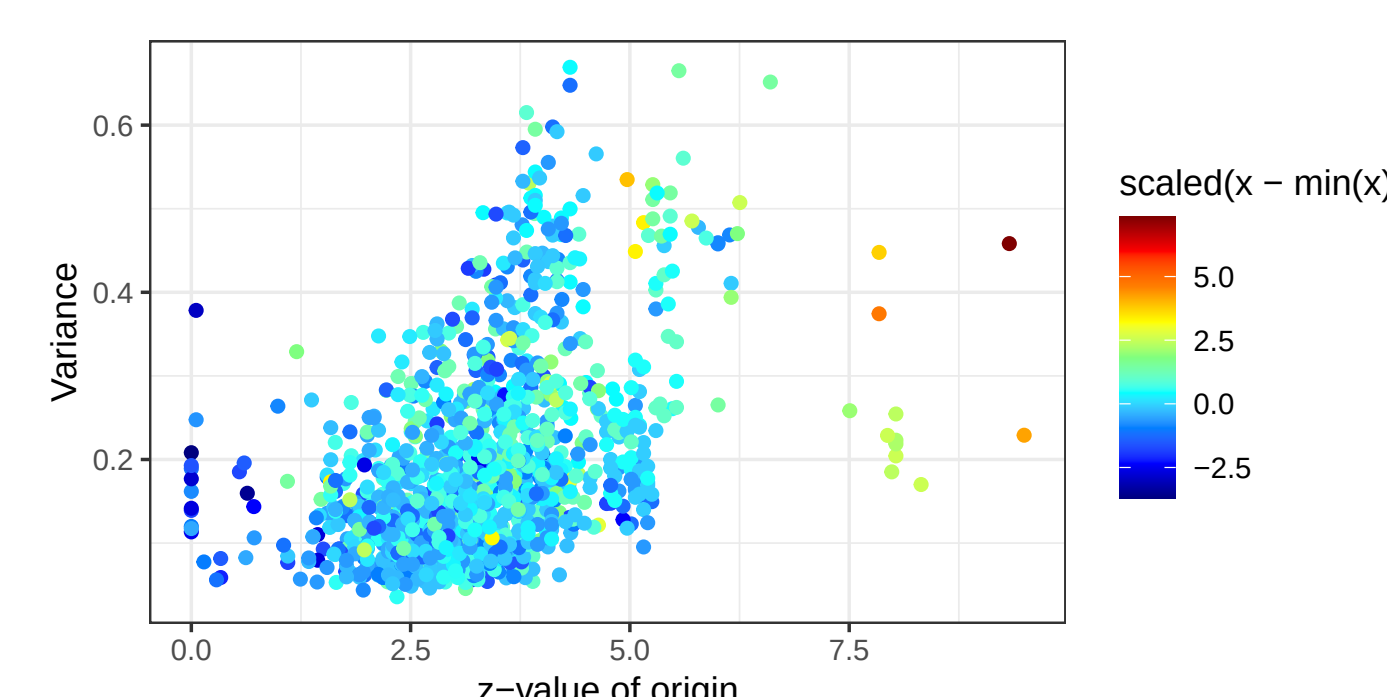


Figure 5: Variance across z-value of origins for all speakers. Corresponding x-values are close to the speaker's minimum x-value when dark blue, farther away when dark red.

Results: tongue grooving in /ɹ/ and /l/

- Using determined origin for each speaker, points on tongue surface were given a θ and r
- Lowest region of tongue surface in cartesian coordinates is tongue root (figure 6, top)
- Lowest region of tongue surface in cylindrical coordinates is location of groove (figure 6, bottom)
- Importantly, lateral edges of tongue are higher relative to center portion of groove
- Tongue grooving for /ɹ/ more posterior than tongue grooving for /l/ (figure 7)
- /ɹ/ productions in figures 6 and 7 are bunched
- /l/ productions in figure 7 are uvular (even in onset)
- No noticeable effects of vowel context on tongue grooving

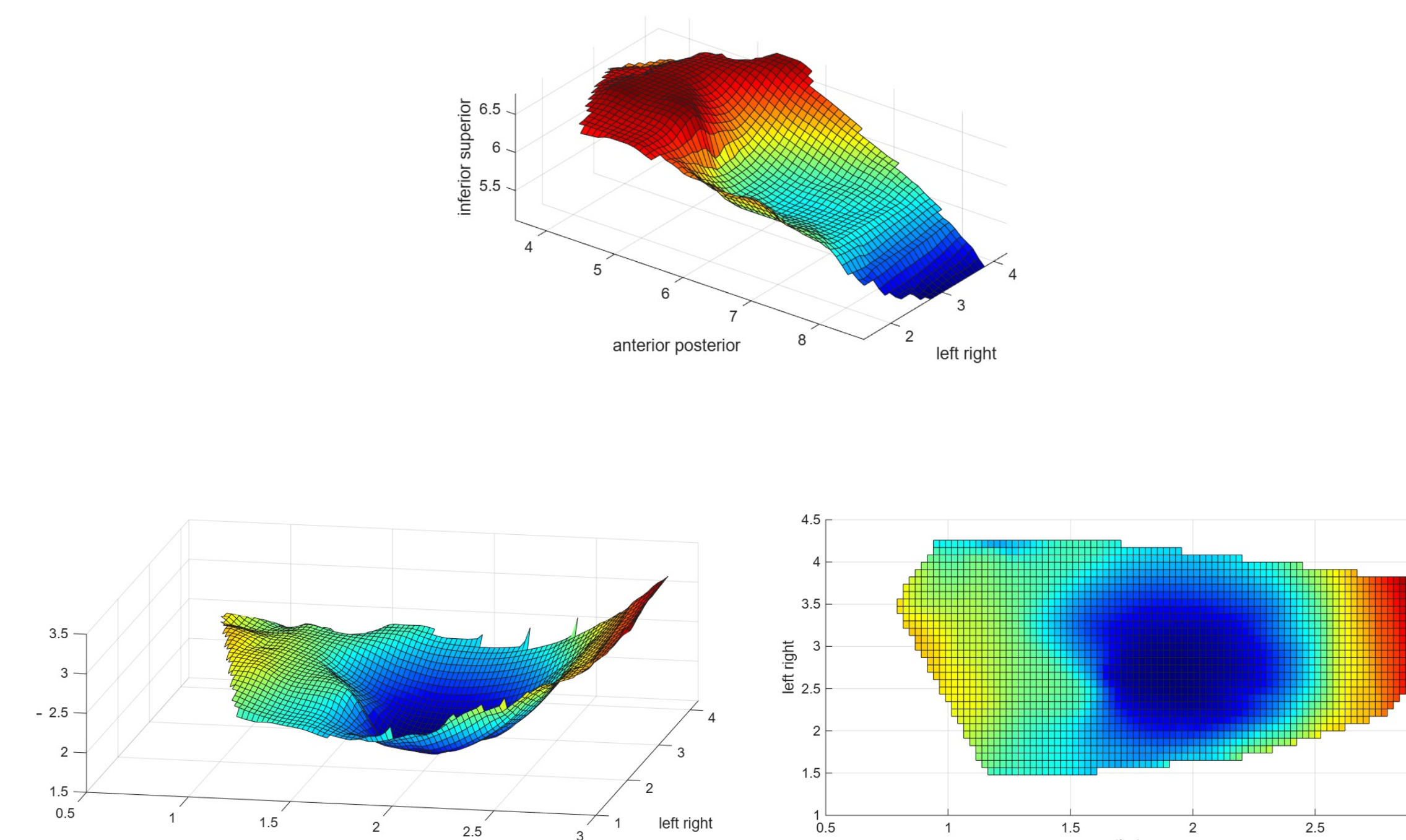


Figure 6: Tongue surface of onset /ɹ/ before i before conversion (top) and after conversion (bottom). The tongue tip is towards the left and the tongue root is towards the right

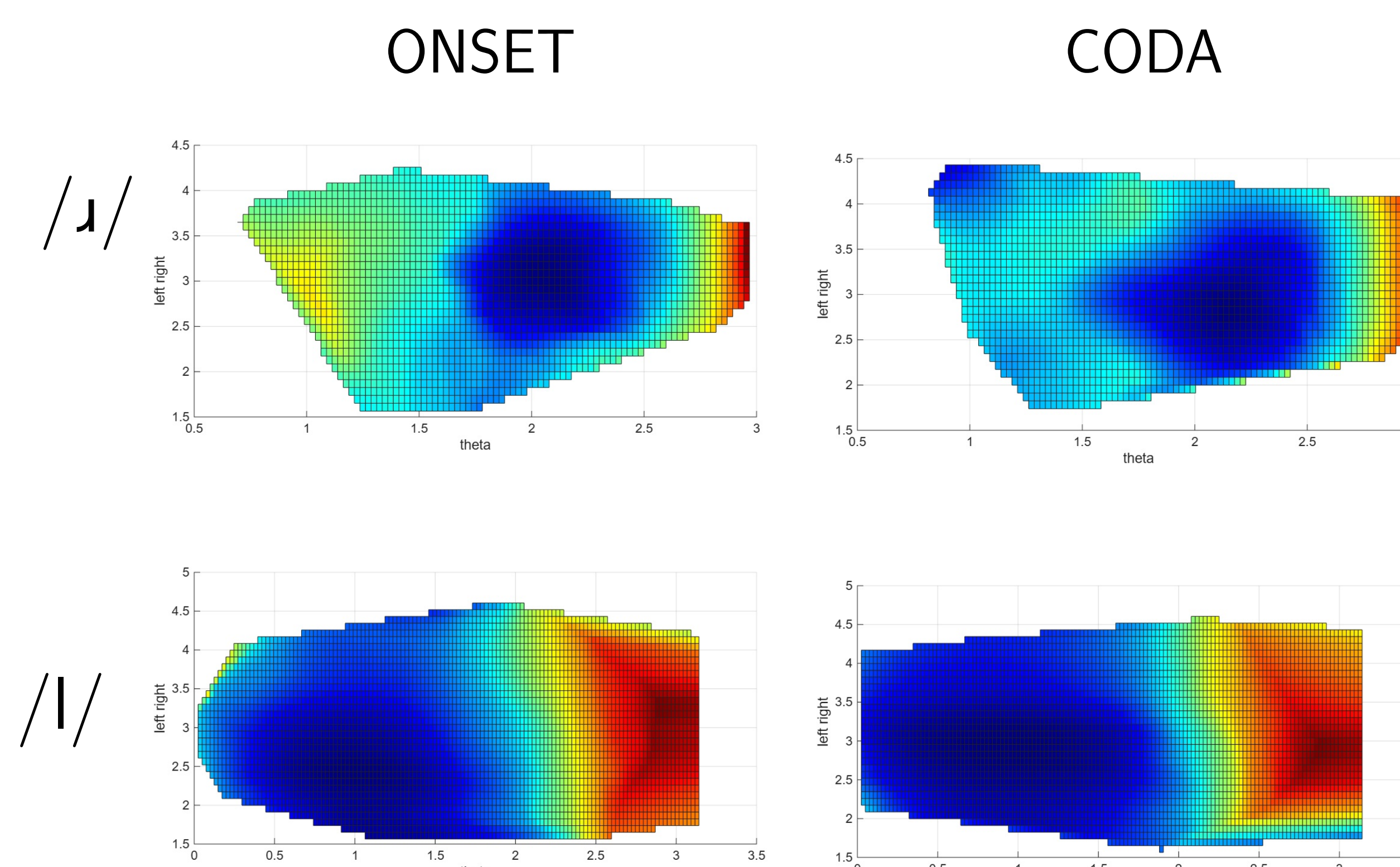


Figure 7: Top down view of tongue surface in cylindrical coordinates, from one speaker. All productions have /a/ as nuclei. Dark blue areas are closest to origin (lowest r-values) and dark red areas are farthest from origin (highest r-values)

Results: tongue grooving (cont.)

- Tongue grooving not present at rest (figure 8, left) nor during vowel productions, such as /ʌ/ (figure 8, right)
- Calculated the global minimum of the tongue surface in cylindrical coordinates
- Contour maps such as figure 9 could be used to automatically detect grooving

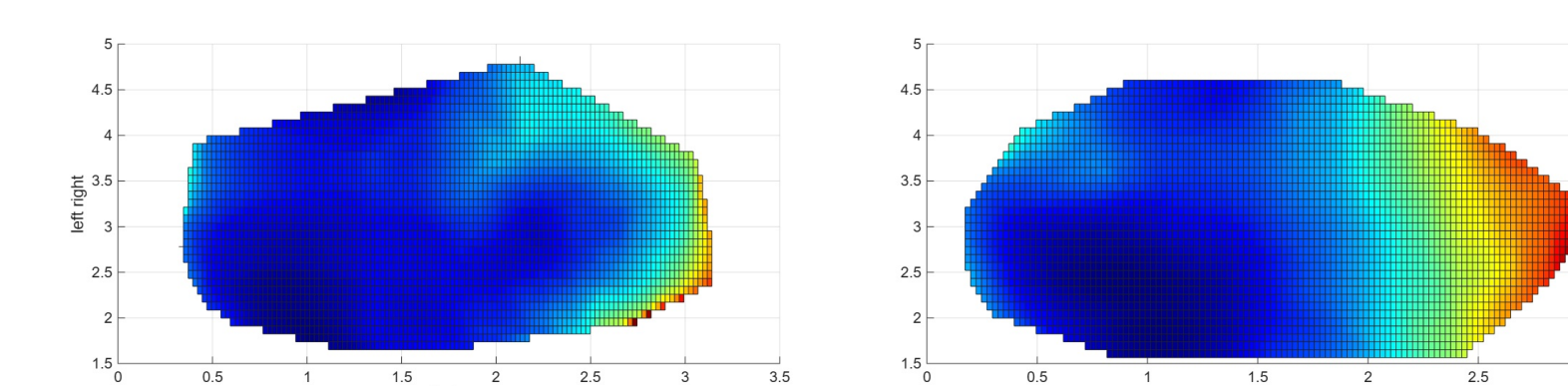


Figure 8: Tongue surface at rest (left) and during /ʌ/ in 'again' (right). No grooving in either

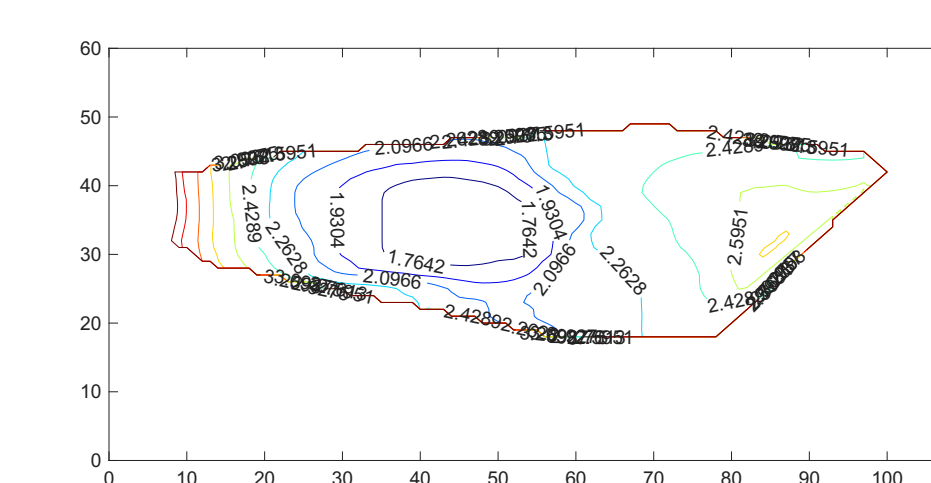


Figure 9: Contour map of /ɹ/, based on calculations of the global minimum. Tongue tip is towards the right and tongue root is towards the left

Conclusions

- Automatic method of determining origin and conversion into cylindrical coordinates works well across speakers
- Bunched /ɹ/ has a posterior groove in onset and coda position
- Grooving in /l/ appears to be more anterior, but less prominent than expected
 - More exploration to be done with comparing pharyngeal vs. uvular/velar productions of /l/
- Grooving was not present at rest or with vowels
 - "Modal" grooving in tongue root
- Calculating global minimum of tongue surfaces could be a way of automatically detecting tongue grooving
 - However, global minima could be result of lateral edges of the tongue being higher, rather than the central portion of the tongue being lower

References

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